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PROJECT REPORT ON

"Hangman Game Using Python"

IN PARTIAL FULFILLMENT OF

BACHELORS OF SCIENCE (AI/ML)

SEMESTER 1– 2025-26

PROJECT GUIDE SANJANA BHANGALE

SUBMITTED BY: PRAPTI GHODEKAR

ROLL NO: 1765

Evaluation sheet for continuous assessment with rubrics

Class: B.Sc. AI/ML
Subject: Hangman game using python

Details about the continuous Assessment 2/Project work

Name of the Student: **Prapti Ghodekar**

Roll Number: **1765**

Class / Division: **B.Sc. AI/ML**

Name of Evaluator: **Prof. Sanjana Bhangale**

Please circle appropriate score

Grading criteria	Fair	Good	Excellent	Total
Introduction/Description of the case				/3
SOWT Analysis of the company used for case analysis pertaining to the case(Strength of CA2 topic : e.g. main important feature of CA2,Weakness :limitation of the project ,Opportunities : in carrier in the future , Threat : obstracles that can cause failure to project CA2)				/5
Learning from the case				/4
Delivery/presentation Skills				/3
Total				15

HANGMAN GAME USING PYTHON

NAME : PRAPTI GHODEKAR

ROLL NO : 1765

CLASS : FYAIML

Hangman is a guessing game where one player thinks of a word, and the other tries to guess it by suggesting letters. For each incorrect letter, a part of a drawing of a hangman is added. The guesser wins by guessing the entire word within the allowed number of incorrect guesses, while the guesser loses if the hangman drawing is completed before the word is solved.

Rules of playing the game

1. **Choose a word:** One player selects a secret word or phrase.
2. **Represent the word:** The selected word is represented by a row of dashes, with each dash standing for a letter in the word.
3. **Guess letters:** The other player guesses letters one by one.
4. **Correct guesses:** If the guessed letter is in the word, the word-thinker fills in all the correct positions in the row of dashes.
5. **Incorrect guesses:** If the guessed letter is not in the word, a part of the hangman drawing is added.
6. **Win or lose:**
    - **Win:** The guessing player wins if they correctly guess the entire word before the hangman drawing is completed.
    - **Lose:** The guessing player loses if the hangman is completely drawn before they guess the word.

We can play this using python functions in which we create a stages of hangman losing it parts one by one. After that we have written a code using import command. We also have create a word.txt file which contains some words which are user going to guess. Further we have written the main codes and then we will run the code and start playing the hangman game.

This game helps us to correctly identify letters to spell out the hidden word, reinforcing their spelling knowledge. Choosing letters to guess a word helps us expand our vocabulary and learn new words. The game encourages active participation and motivation, making learning more engaging for students. There is no such disadvantages of this game but its potential is to be boring and unengaging, its limited educational value, the problematic imagery of death and lynching.

This game is not harmful to anyone as it provides lots of knowledge and it is very easy to play.

- Words are stored in the words.txt file

```
programming
machine
python
management
pandas
human
college
```

CODE :

```
# Python program to implement Hangman Game

stages = [r'''
   +---+
   |   |
   O   |
  /|\  |
  / \  |
       |
=========
''', r'''
   +---+
   |   |
   O   |
  /|\  |
  /    |
       |
=========
''', r'''
   +---+
   |   |
   O   |
  /|\  |
       |
       |
=========
''', r'''
   +---+
   |   |
   O   |
  /|   |
       |
       |
=========
```

```
''', r'''
  +---+
  |   |
  O   |
  |   |
      |
      |
=========
''', r'''
  +---+
  |   |
  O   |
      |
      |
      |
=========
''', r'''
  +---+
  |   |
      |
      |
      |
      |
=========
''']
```

```python
import random

# Function to randomly select a word from dictionary
def get_word():
    try:
        with open('words.txt', 'r') as f:
            words_list = f.read().splitlines()
        return random.choice(words_list)
    except FileNotFoundError:
        print("words.txt not found. Please add a word list file in the same folder.")
        exit()

def hangman():
    chosen_word = get_word()
    display = ["_"] * len(chosen_word)
    guessed_letters = set()  # new set to track guessed letters

    end_of_loop = False
    lives = 6

    print("\n-------------Welcome to Hangman-------------\n")
    print("Guess the word:- ", ' '.join(display))
    print(f"Lives: {lives}")

    while not end_of_loop:
        guess = input("Guess a Letter: ").lower()

        # check if letter was already guessed
        if guess in guessed_letters:
            print(f"You already guessed '{guess}'. Try a different letter.")
            continue
        else:
            guessed_letters.add(guess)

        if guess not in chosen_word:
            lives -= 1
```

```python
        for index, char in enumerate(chosen_word):
            if char == guess:
                display[index] = guess

        print(' '.join(display))
        print(f"Lives: {lives}")
        print(stages[6 - lives])

        if "_" not in display:
            print("You Win!")
            end_of_loop = True

        if lives == 0:
            print("You Lose")
            print(f"The word was: {chosen_word}")
            end_of_loop = True

# Main loop
while True:
    ask = input("Do you want to play Hangman? (y/n): ").lower()
    if ask in ('y', 'yes'):
        hangman()
    elif ask in ('n', 'no'):
        print("Program Exit Successful")
        break
    else:
        print("Invalid input. Please enter 'y' or 'n'.")
```

OUTPUT :

```
========================= RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python313/game hungman.py =======
Do you want to play Hangman? (y/n): y

-------------Welcome to Hangman-------------

Guess the word:-  _ _ _ _ _ _ _
Lives: 6
Guess a Letter: c
_ _ c _ _ _ _
Lives: 6

  +---+
  |   |
  o   |
 /|\  |
 / \  |
      |
=========

Guess a Letter: a
_ a c _ _ _ _
Lives: 6

  +---+
  |   |
  o   |
 /|\  |
 / \  |
      |
=========

Guess a Letter: l
_ a c _ _ _ _
Lives: 5
```

```
  +---+
  |   |
  O   |
 /|\  |
 /    |
      |
=========

Guess a Letter: m
m a c _ _ _ _
Lives: 5

  +---+
  |   |
  O   |
 /|\  |
 /    |
      |
=========

Guess a Letter: h
m a c h _ _ _
Lives: 5

  +---+
  |   |
  O   |
 /|\  |
 /    |
      |
=========
```

```
Guess a Letter: i
m a c h i _ _
Lives: 5

  +---+
  |   |
  O   |
 /|\  |
 /    |
      |
=========

Guess a Letter: n
m a c h i n _
Lives: 5

  +---+
  |   |
  O   |
 /|\  |
 /    |
      |
=========

Guess a Letter: e
m a c h i n e
Lives: 5

  +---+
  |   |
  O   |
 /|\  |
 /    |
      |
=========
```

```
You Win!
Do you want to play Hangman? (y/n): n
Program Exit Successful
```

```
Do you want to play Hangman? (y/n): Y

-------------Welcome to Hangman-------------

Guess the word:-  _ _ _ _ _ _
Lives: 6
Guess a Letter: P
_ _ _ _ _ _
Lives: 5

  +---+
  |   |
  O   |
 /|\  |
 /    |
      |
=========

Guess a Letter: D
_ _ _ _ _ _
Lives: 4

  +---+
  |   |
  O   |
 /|\  |
      |
      |
=========
```

```
_ _ _ _ _ _
Lives: 3

  +---+
  |   |
  O   |
 /|   |
      |
      |
=========

Guess a Letter: L
_ _ _ _ _ _
Lives: 2

  +---+
  |   |
  O   |
  |   |
      |
      |
=========

Guess a Letter: L
You already guessed 'l'. Try a different letter.
Guess a Letter: O
_ _ _ _ _ _
Lives: 1
```

```
  +---+
  |   |
  O   |
      |
      |
      |
=========

Guess a Letter: P
You already guessed 'p'. Try a different letter.
Guess a Letter: T
_ _ _ _ _ _
Lives: 0

  +---+
  |   |
      |
      |
      |
      |
=========

You Lose
The word was: INDIAN
```

```
Do you want to play Hangman? (y/n): n
Program Exit Successful
```